

Task 3: Car Price Prediction

Objective

Predict car selling prices using Linear Regression after preprocessing the dataset.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score

df = pd.read_csv("car_data.csv")
display(df.head())
display(df.describe())
print(df.info())
print(df.isnull().sum())

df["Car_Age"] = 2025 - df["Year"]
df.drop(["Car_Name", "Year"], axis=1, inplace=True)

le = LabelEncoder()
for col in ["Fuel_Type", "Selling_type", "Transmission"]:
    df[col] = le.fit_transform(df[col])

X = df.drop("Selling_Price", axis=1)
y = df["Selling_Price"]

X_train, X_test, y_train, y_test =
train_test_split(X, y, test_size=0.2, random_state=42)

model = LinearRegression()
model.fit(X_train, y_train)
pred = model.predict(X_test)

mse = mean_squared_error(y_test, pred)
print("MAE:", mean_absolute_error(y_test, pred))
print("MSE:", mse)
print("RMSE:", np.sqrt(mse))
print("R2:", r2_score(y_test, pred))

plt.figure(figsize=(6,6))
plt.scatter(y_test, pred)
```

```
plt.xlabel("Actual")  
plt.ylabel("Predicted")  
plt.title("Actual vs Predicted Selling Price")  
plt.show()
```

Conclusion

The regression model predicts selling prices with evaluation using MAE, MSE, RMSE and R^2 score.